

CIS326: IT Infrastructure Management Project Report.
Network Infrastructure for Eastern Province Public Libraries

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1. Introduction

This research project investigates the implementation and performance of dynamic routing protocols in small-scale networks, focusing on a comparative analysis of Routing Information Protocol (RIP) and Open Shortest Path First (OSPF). The objective is to assess each protocol's behavior regarding convergence time, network scalability, and packet loss during simulated outages. Additionally, the study evaluates Cisco Packet Tracer—used as the primary simulation tool—based on its CPU and memory consumption, as well as simulation efficiency..

2. Methodology

The project was structured into two primary stages:

- Implementation and testing of RIP using a simple three-router topology.
- Deployment and assessment of OSPF using a comparable topology with minor optimizations to reflect protocol-specific requirements.

Each topology included end-user devices (PCs), IPPhones and intermediary switches, representing segmented networks. The process involved:

- Assigning private IP addresses to relevant interfaces.
- Enabling dynamic routing protocols and defining routing advertisements.
- Validating configurations using commands like `show`, `debug`, `ping`, and `traceroute`.
- Simulating internet access via default routes.
- Observing routing behavior during artificial link disruptions.

Cisco Packet Tracer served as the simulation platform, providing a controlled environment to build, test, and visualize network designs while gathering performance data.

For the protocol comparison, packet loss was calculated using the following formula:

$$\text{Packet Loss (\%)} = \frac{\text{Sent Packets} - \text{Received Packets}}{\text{Sent Packets}} \times 100$$

Common causes of packet loss considered during testing included:

1. Excessive network traffic leading to congestion.
2. Packet collisions on shared network segments.
3. Physical layer issues.
4. Receiver-side buffer overflows.

3. Proposed Model

The topology for both RIP and OSPF featured:

- Three routers (Router1, Router2, and ISP-R1).
- PCs and IPphones connected through intermediary switches.
- Serial point-to-point links between routers, using /30 subnets.

RIP Topology

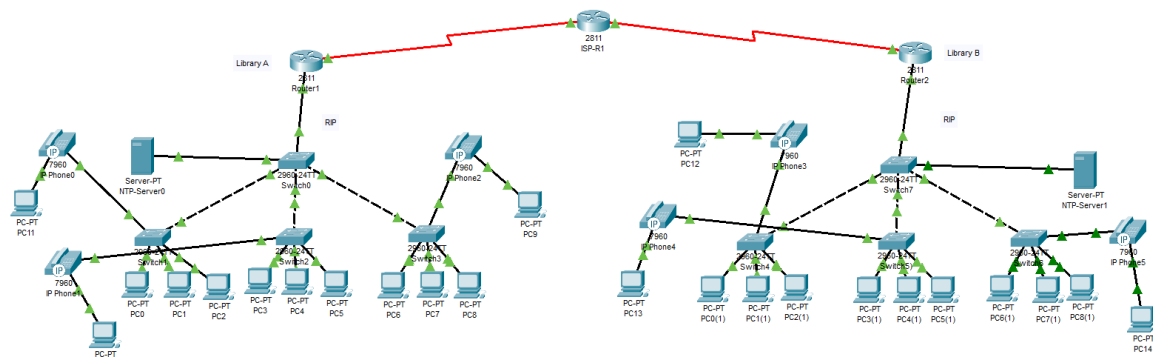


Figure 1: RIP Topology

Addressing Table

Table 1: the addressing table for the RIP network

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.10.1	255.255.255.224	
	S1/0	10.10.10.1	255.255.255.252	
	Fa0/0.10	192.168.10.161	255.255.255.224	
	Fa0/0.20	192.168.10.193	255.255.255.224	
R2	Fa0/0	192.168.10.97	255.255.255.224	
	S1/1	20.20.20.1	255.255.255.252	
	Fa0/0.10	192.168.10.33	255.255.255.224	
	Fa0/0.20	192.168.10.65	255.255.255.224	
ISP-R1	S1/0	10.10.10.2	255.255.255.0	

	S1/1	20.20.20.2	255.255.255.252	
NTP-Server0	NIC	192.168.10.162	255.255.255.224	192.168.10.161
NTP-Server1	NIC	192.168.10.34	255.255.255.224	192.168.10.33
Library-A-PCs	NIC	DHCP	255.255.255.224	192.168.10.202
Library-B-PCs	NIC	DHCP	255.255.255.224	192.168.10.65

OSPF topology

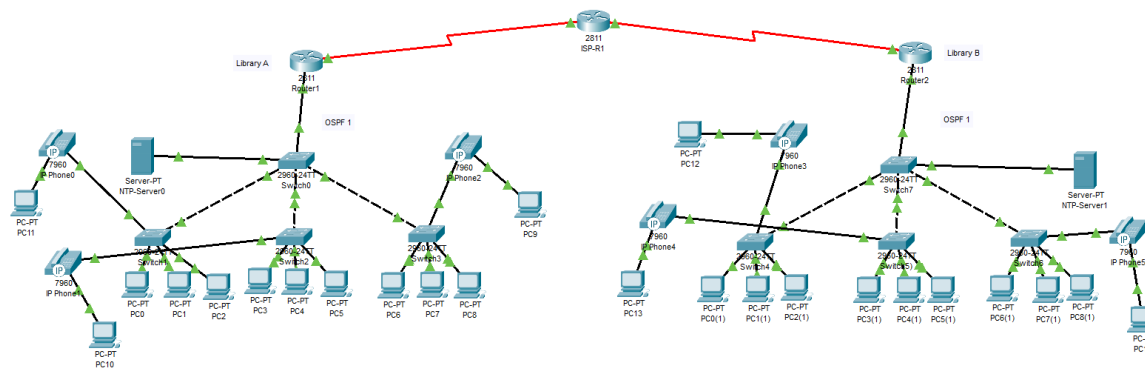


Figure 2: OSPF topology

Addressing Table

Table 2: the addressing table for the OSPF network

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	192.168.10.1	255.255.255.224	
	S1/0	10.10.10.1	255.255.255.252	
	Fa0/0.10	192.168.10.161	255.255.255.224	
	Fa0/0.20	192.168.10.193	255.255.255.224	
R2	Fa0/0	192.168.10.97	255.255.255.224	
	S1/1	20.20.20.1	255.255.255.252	
	Fa0/0.10	192.168.10.33	255.255.255.224	
	Fa0/0.20	192.168.10.65	255.255.255.224	

ISP-R1	S1/0	10.10.10.2	255.255.255.0	
	S1/1	20.20.20.2	255.255.255.252	
NTP-Server0	NIC	192.168.10.162	255.255.255.224	192.168.10.161
NTP-Server1	NIC	192.168.10.34	255.255.255.224	192.168.10.33
Library-A-PCs	NIC	DHCP	255.255.255.224	192.168.10.202
Library-B-PCs	NIC	DHCP	255.255.255.224	192.168.10.65

4. Result and Discussion

4.1 RIP vs OSPF – Protocol Comparison

A structured evaluation of RIP and OSPF was conducted, highlighting key functional distinctions:. Here's a summary of key differences:

Table 3: RIP vs OSPF – Protocol Comparison

Feature	RIP	OSPF
Type	Distance Vector	Link-State
Algorithm	Bellman-Ford	Dijkstra's Shortest Path First
Metric	Hop Count (Max 15 hops)	Cost (Bandwidth-based)
Convergence Speed	Slow	Fast
Updates	Periodic (30s)	Triggered
Scalability	Low	High
Authentication Support	Limited	Strong (e.g., MD5)
Route Summarization	Manual	Automatic & Manual
Administrative Distance	120	110
VLSM Support	RIP v2 Only	Fully Supported
Multicast Address	224.0.0.9 (RIPv2)	224.0.0.5 / 224.0.0.6

4.2 Packet loss

To quantify protocol performance, packet loss was evaluated using the `ping` utility between PCs in separate subnets following initial convergence. Using the same formula mentioned earlier, the estimated packet loss was calculated based on convergence delays.

As mentioned earlier to calculate the packet loss using the following equation:

$$\text{Packet Loss (\%)} = \frac{\text{Sent Packets} - \text{Received Packets}}{\text{Sent Packets}} \times 100$$

$$\text{Packet loss} = \frac{(4 - 4)}{4} \times 100 = 0\%$$

RIP Packet Loss

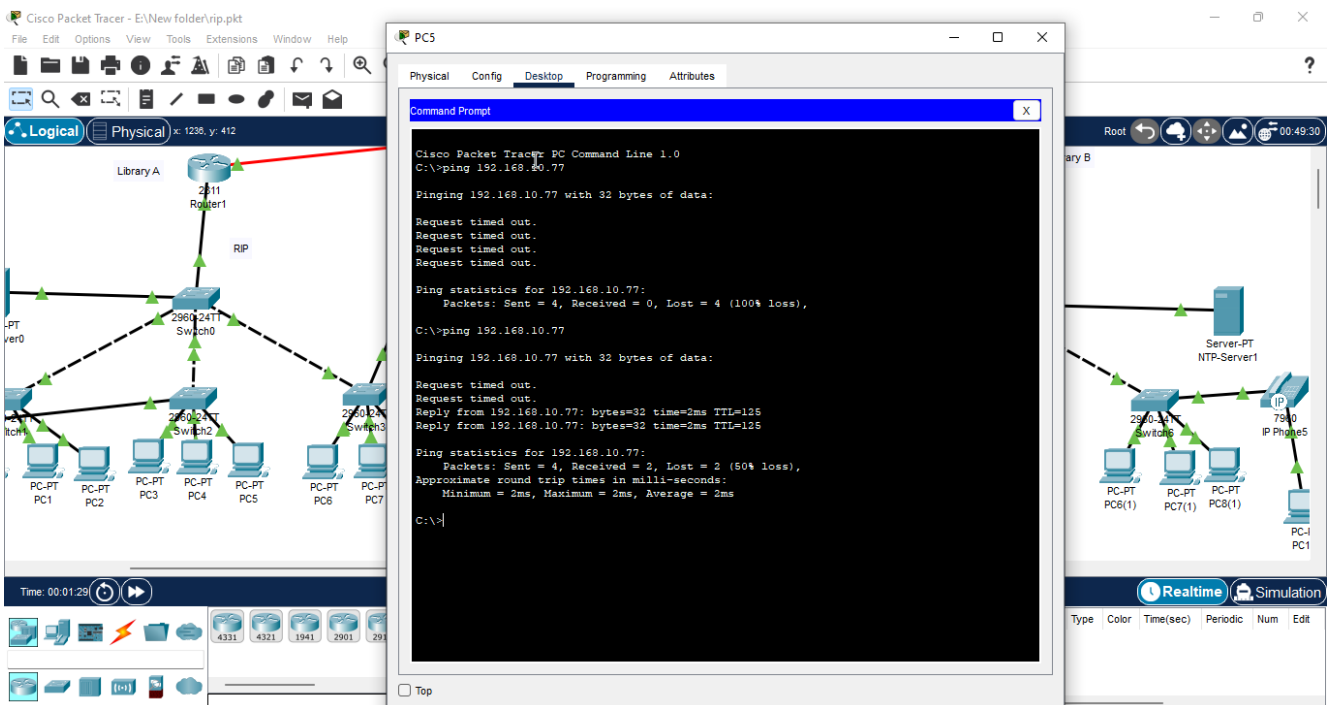


Figure 3: RIP Packet Lost

$$\text{Packet Loss (\%)} = \frac{4 - 0}{4} \times 100 = 100\%$$

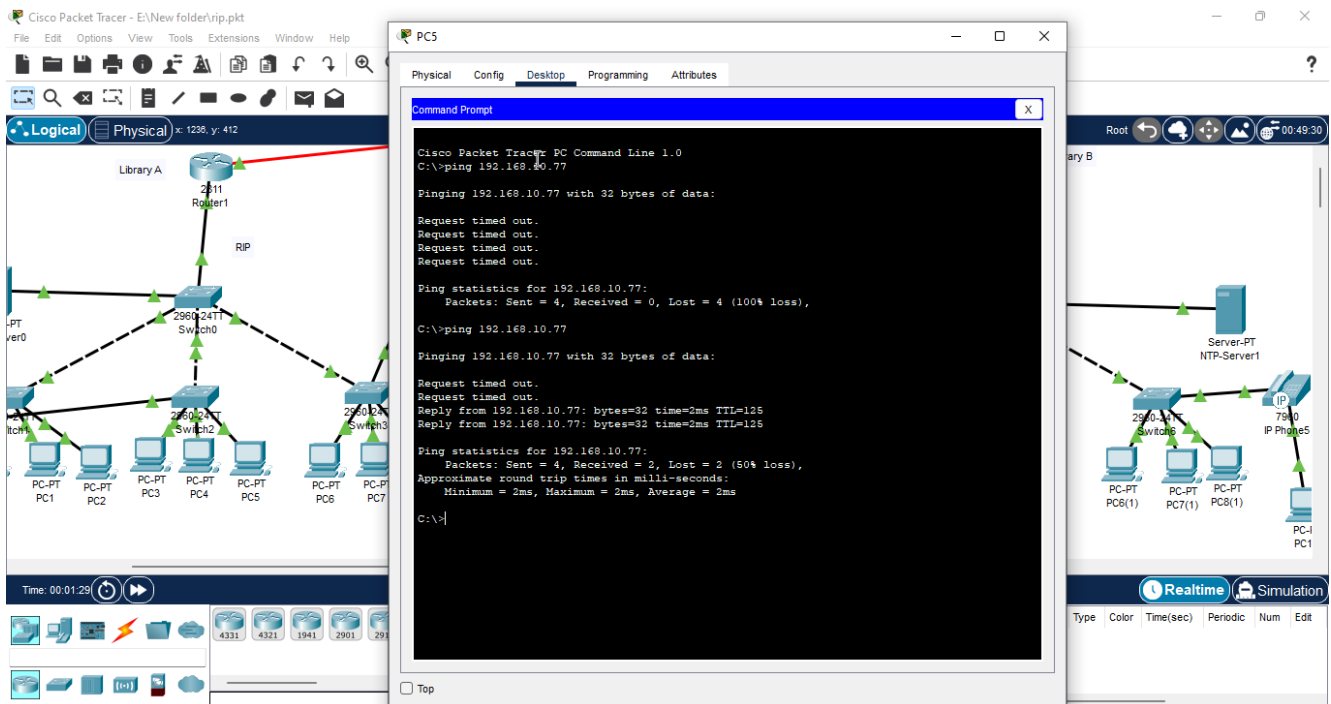


Figure 4:RIP Packet Lost

$$\text{Packet Loss (\%)} = \frac{4-2}{4} \times 100 = 50\%$$

OSPF Packet Loss

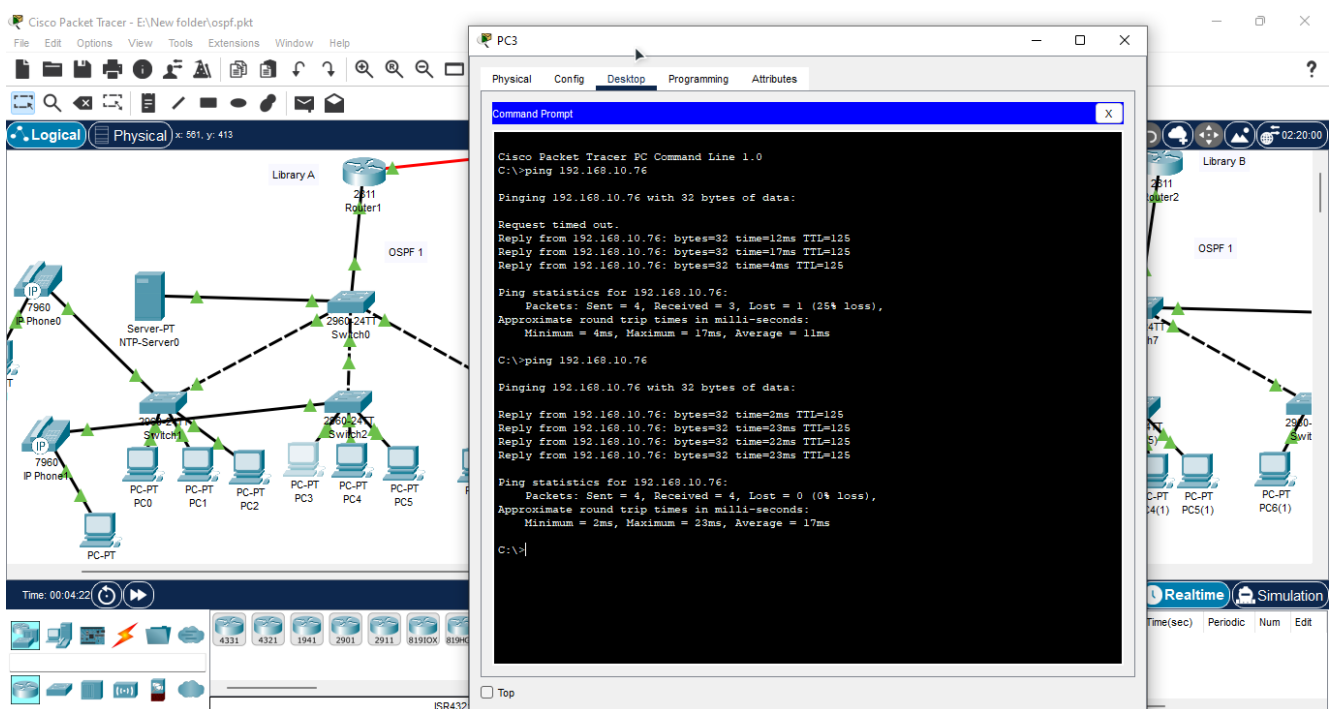


Figure 5: OSPF Packet Lost

$$\text{Packet Loss (\%)} = \frac{4-3}{4} \times 100 = 25\%$$

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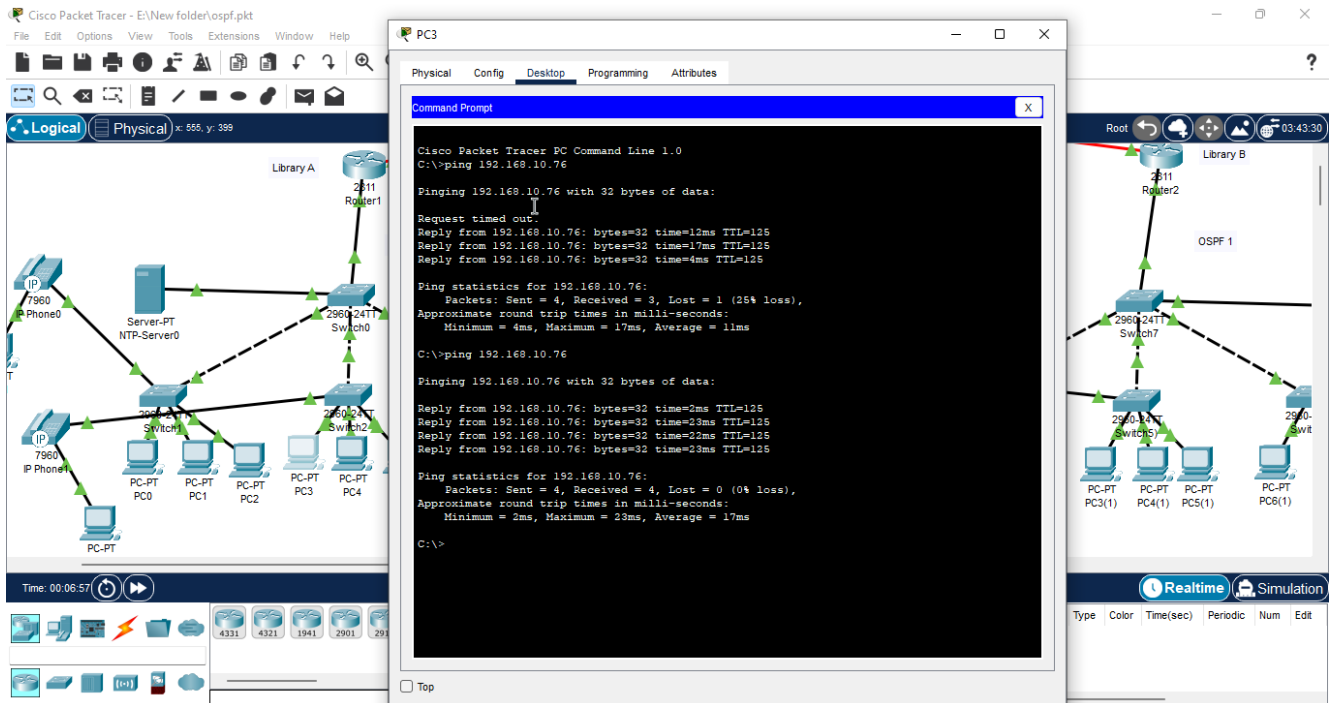


Figure 6: OSPF Packet Lost

$$\text{Packet Loss (\%)} = \frac{4-4}{4} \times 100 = 0\%$$

Estimated Performance:

RIP

- Convergence Time: ~60 seconds
- Estimated Packet Loss: $1,000 \times 60 = 60,000$ packets

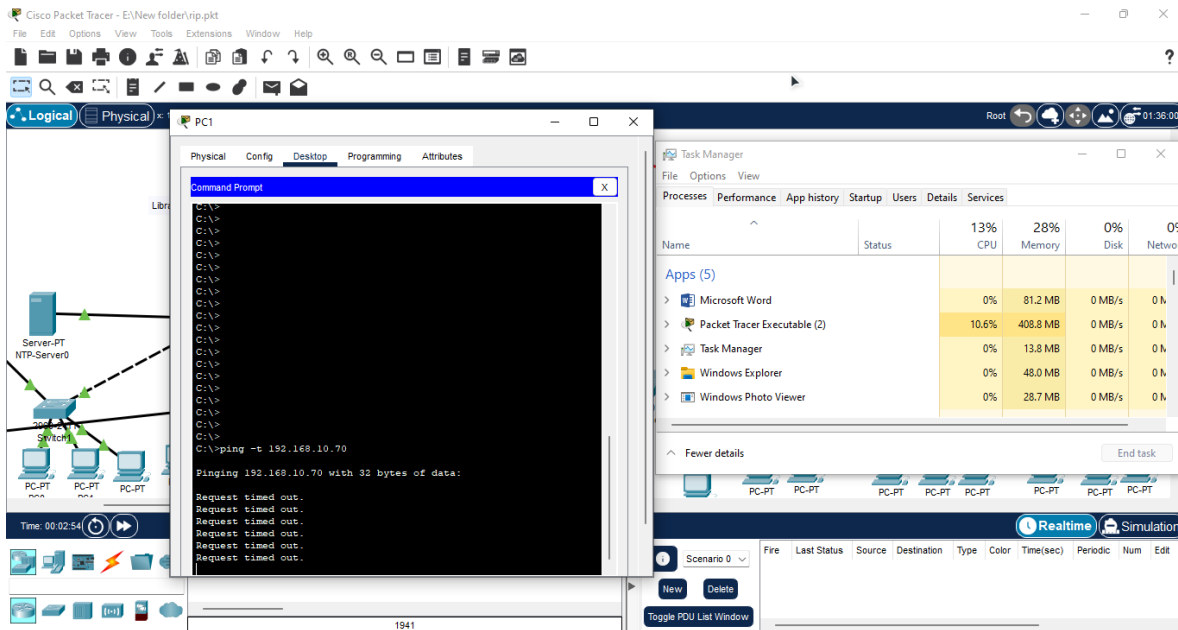


Figure 7: RIP convergence time Packet Loss

OSPF

- Convergence Time: ~5 seconds
- Estimated Packet Loss: $1,000 \times 5 = 5,000$ packets

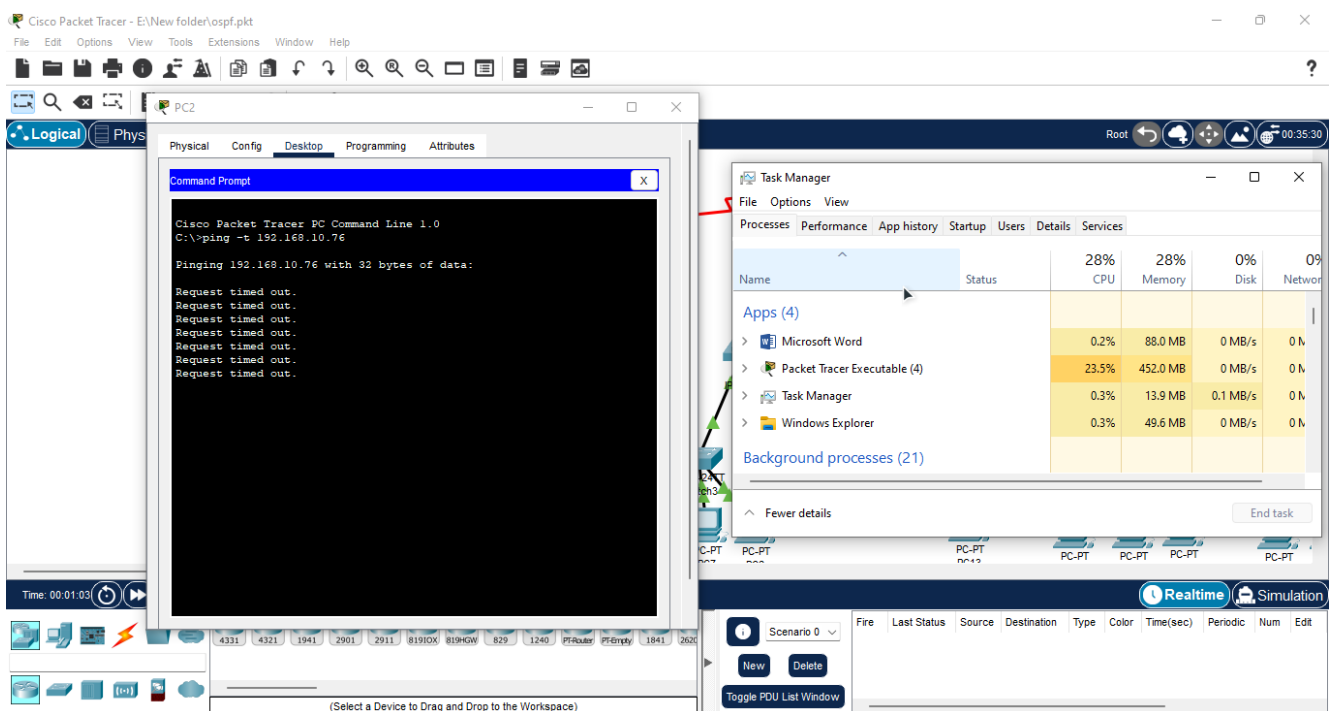


Figure 8: OSPF convergence time Packet Loss

This demonstrates that OSPF's rapid convergence results in significantly lower packet loss, reinforcing its suitability for networks where stability and uptime are critical..

Evaluation of Cisco Packet Tracer

As part of the project, Cisco Packet Tracer's performance was evaluated to determine its effectiveness as a simulation tool under various network conditions. The assessment focused on three primary system resource metrics:

1. CPU Utilization

Cisco Packet Tracer demonstrated efficient CPU usage across different simulation scales:

- For basic or idle networks, CPU consumption remained minimal, typically between **5% and 10%**.
- When simulating moderate network environments with approximately 20 to 30 active devices, CPU usage increased to a range of **10% to 30%**.
- In more demanding scenarios, involving dynamic protocols such as OSPF with continuous traffic, utilization occasionally spiked up to **40% to 60%**, reflecting the increased computational demand during real-time protocol execution.

2. Memory Usage

Memory usage was directly proportional to the complexity of the simulated topology:

- Smaller configurations consumed approximately **250–400 MB** of RAM.
- Mid-sized projects with more devices and active routing processes typically required **500–800 MB** of RAM.

The screenshot displays the Cisco Packet Tracer interface with a network topology and the Windows Task Manager window open.

Network Topology:

- Library A:** Contains a 2611 Router1.
- Server-PT:** NTP-Server0.
- Switches:** 2960-24TT Switch0, 2960-24TT Switch1, 2960-24TT Switch2, 2960-24TT Switch3.
- PCs:** PC-PT PC11, PC-PT PC3, PC-PT PC4, PC-PT PC5, PC-PT PC6, PC-PT PC7, PC-PT PC8, PC-PT PC9, PC-PT PC10, PC-PT PC12.
- IP Phones:** IP Phone0, IP Phone1.
- Connections:** Router1 is connected to Switch0. Switch0 is connected to Switch1 and Switch2. Switch1 is connected to Switch3. Switch2 is connected to Switch3. Switch1 is connected to PC11 and IP Phone1. Switch2 is connected to PC3, PC4, and PC5. Switch3 is connected to PC6, PC7, PC8, PC9, and PC10.

Windows Task Manager:

The Task Manager window shows the following processes:

Name	Status	CPU	Memory	Disk	Network
Apps (3)					
Packet Tracer Executable (2)		2.6%	380.0 MB	0 MB/s	0 M
Task Manager		0.8%	13.9 MB	0 MB/s	0 M
Windows Explorer		0.3%	44.2 MB	0 MB/s	0 M
Background processes (23)					
Agent Service (32 bit)		0%	6.0 MB	0 MB/s	0 M
Antimalware service protection ...		0%	2.3 MB	0 MB/s	0 M
Application Host Service (32 bit)		0.3%	50.7 MB	0.1 MB/s	0 M
AV Console (32 bit)		0.2%	2.6 MB	0 MB/s	0 M

The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with a 'Server-PT' connected to a central 'Switch'. Three 'PC-PT' devices are connected to the switch. A 'Command Prompt' window is open on one of the PCs, showing a series of ping commands and successful replies from IP address 192.168.10.70 to 192.168.10.125. To the right, the 'Task Manager' application is open, listing running processes such as Microsoft Word, Packet Tracer Executable (2), Task Manager, Windows Explorer, and Windows Photo Viewer. The bottom status bar indicates the simulation is running in 'Realtime' mode.

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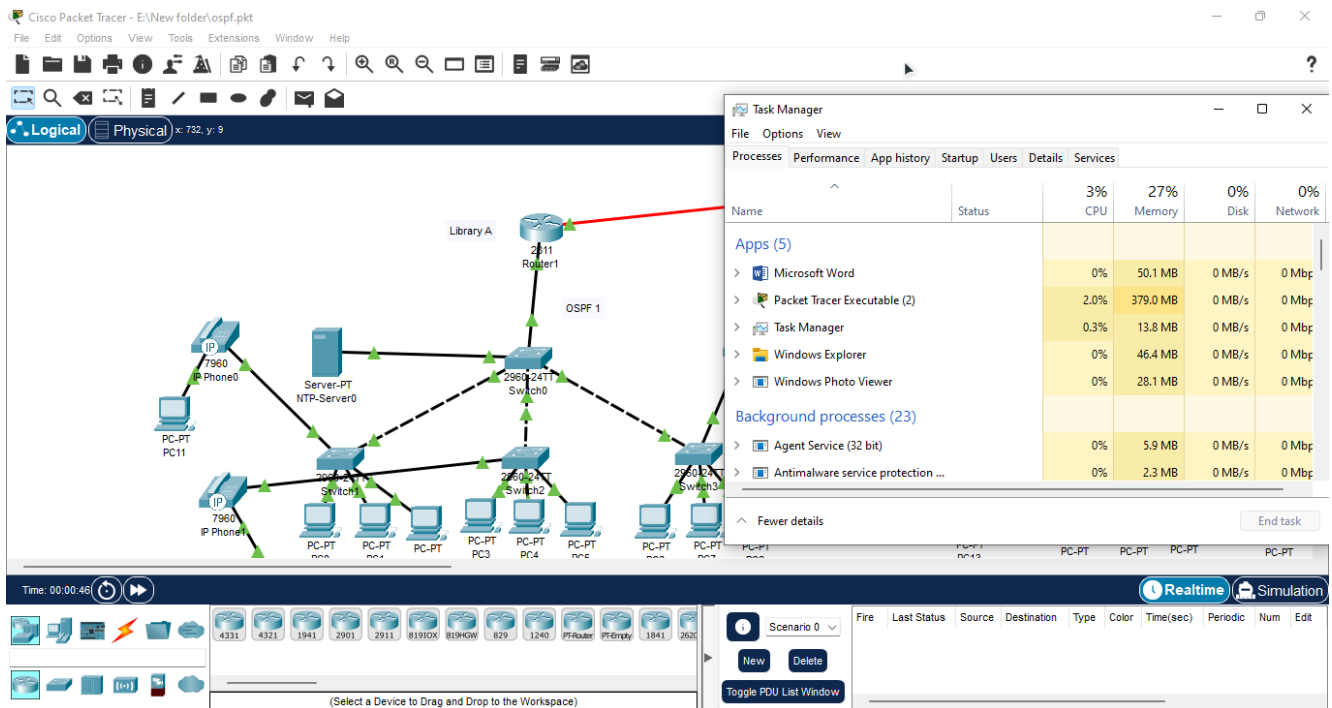


Figure 11:OSPF idle state

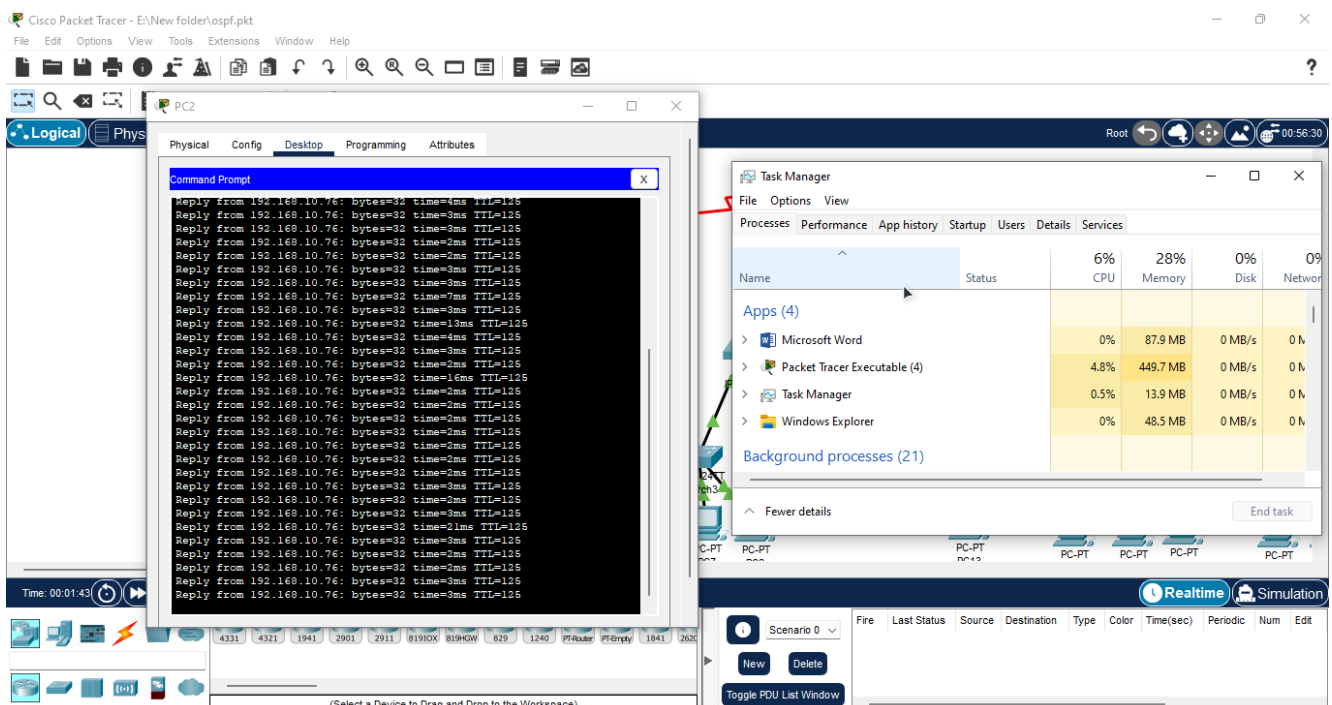


Figure 12: OSPF active state

5. Conclusion

The investigation successfully demonstrated key contrasts between RIP and OSPF in terms of design complexity, convergence efficiency, and overall network performance. While RIP remains a straightforward choice for smaller or static environments, its scalability limitations and slower response during link failures reduce its viability for more dynamic use cases.

In contrast, OSPF exhibited superior adaptability, rapid convergence, and better control over routing advertisements—qualities essential for modern enterprise networks. Its support for hierarchical design and robust authentication mechanisms further strengthen its suitability for large-scale deployments.

Cisco Packet Tracer proved to be a valuable educational and research tool, effectively simulating routing environments while maintaining reasonable system resource demands. Its visual feedback and diagnostic tools provided insightful data that enhanced understanding of routing behaviors and network dynamics under various operational loads.

6. References

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